

**ENHANCING THE COMPETENCY OF USING PROBLEM
SOLVING STRATEGY IN MATHEMATICS AMONG
SENIOR SECONDARY TEACHERS IN SRI LANKA**

BY

NAVARAJAH PRASHANTHAN

FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY

FACULTY OF EDUCATION

UNIVERSITY OF COLOMBO

SRI LANKA

2027

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**NAVARAJAH PRASHANTHAN
(BSc, BEd, MSc in Science Education, MPhil)**

*in the partial fulfilment of the requirement
for the award of the degree of*

DOCTOR OF PHILOSOPHY

of the

**FACULTY OF EDUCATION
UNIVERSITY OF COLOMBO
SRI LANKA.**

2027

DECLARATION

I do hereby declare that the work reported in this thesis was exclusively carried out by me under the supervision of Dr. D.V. K.P. Seneviratne. It describes the results of my own independent research except where due reference has been made in the text. No part of this thesis has been submitted earlier or concurrently for the same or any other degree.

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Candidate to draft. Include the generative-AI use declaration required by the UoC GenAI policy, built from ai-use-log.md.

ABSTRACT

Enhancing the competency of using problem solving strategy in mathematics among senior secondary teachers in Sri Lanka

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Candidate to draft (purpose, design, sample, instruments, analysis, key findings).

Keywords: [keyword], [keyword].

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LIST OF ABBREVIATIONS

CFA	Confirmatory Factor Analysis
CVI	Content Validity Index
DOE	Department of Examinations
EFA	Exploratory Factor Analysis
G.C.E (O/L)	General Certificate of Education (Ordinary Level)
MKT	Mathematical Knowledge for Teaching
MOE	Ministry of Education
NCTM	National Council of Teachers of Mathematics
NEC	National Education Commission
NIE	National Institute of Education

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Mathematics education plays a crucial role in developing logical reasoning, analytical thinking, and problem-solving abilities among students. In the context of secondary education in Sri Lanka, mathematics is a core subject that significantly influences students' academic and professional pathways. However, despite its importance, many students demonstrate difficulties in understanding mathematical concepts and applying them to solve problems effectively.

One of the major contributing factors to this issue is the teaching approach adopted in classrooms. Traditional teaching methods in mathematics often emphasize procedural knowledge and rote learning rather than conceptual understanding and problem-solving skills. As a result, students may perform well in routine exercises but struggle when faced with unfamiliar or complex problems.

Problem-solving strategies are widely recognized as essential tools in mathematics education. These strategies include understanding the problem, devising a plan, carrying out the plan, and reflecting on the solution—principles popularized by George Pólya. The effective use of such strategies helps learners develop deeper understanding and adaptability in mathematics.

Senior secondary teachers play a vital role in fostering these skills. Their competency in applying problem-solving strategies directly impacts students' learning outcomes. However, there is evidence that many teachers lack adequate training, resources, or confidence to integrate these strategies effectively into their teaching practices.

Therefore, enhancing teachers' competency in using problem-solving strategies is essential to improve the quality of mathematics education in Sri Lanka.

1.2 Statement of the Problem

Despite the recognized importance of problem-solving in mathematics, many senior secondary students in Sri Lanka exhibit low performance in solving non-routine mathematical

problems. This issue may stem from teachers' limited use of effective problem-solving strategies during instruction.

Teachers often rely on traditional, teacher-centered methods that focus on memorization and repetitive practice. Consequently, students are not adequately exposed to heuristic approaches or critical thinking processes required for solving complex problems. The problem addressed in this study is the insufficient competency among senior secondary mathematics teachers in Sri Lanka in utilizing problem-solving strategies effectively in their teaching practices.

1.3 Purpose of the Study

The main purpose of this study is to enhance the competency of senior secondary mathematics teachers in Sri Lanka in using problem-solving strategies in their teaching.

1.4 Objectives of the Study

- I. Identify the current level of cognitive abilities of secondary mathematics teachers regarding problem-solving strategies.
- II. Explore the level of use of problem solving strategies in mathematics by senior secondary teachers.
- III. Find out areas for improvement in problem-solving strategy among senior secondary teachers.
- IV. Design and implement an intervention to enhance teachers' competency in applying problem-solving strategies in mathematics.
- V. Assess the progress of use of problem-solving strategies in mathematics after intervention.
- VI. Propose a teacher education module on effective use of problem solving strategies in mathematics in senior secondary education.

1.5 Research Questions

The study is guided by the following research questions:

1. What is the current level of competency of senior secondary mathematics teachers in using problem-solving strategies?
2. What challenges do teachers face in implementing these strategies?
3. How can teachers' competency in using problem-solving strategies be enhanced?

4. What is the impact of the intervention on teachers' instructional practices?

1.6 Significance of the Study

This study is significant for several reasons:

- **Teachers:** It will help teachers improve their instructional strategies and professional competence.
- **Students:** Improved teaching methods will enhance students' problem-solving skills and overall achievement in mathematics.
- **Educational Authorities:** Findings can inform curriculum development and teacher training programs in Sri Lanka.
- **Researchers:** The study will contribute to existing literature on mathematics education and teacher development.

1.7 Scope of the Study

This study focuses on senior secondary mathematics teachers in Sri Lanka. It specifically examines their competency in using problem-solving strategies in classroom instruction. The study may be limited to selected schools or regions due to time and resource constraints.

1.8 Limitations of the Study

The study may face the following limitations:

- Limited sample size, which may affect generalizability.
- Time constraints in implementing and evaluating the intervention.
- Possible bias in self-reported data from teachers.
- Variations in school environments and resources.

1.9 Definition of Key Terms

- **Problem-Solving Strategy:** A systematic approach used to understand and solve mathematical problems (e.g., strategies proposed by George Pólya).
- **Competency:** The combination of knowledge, skills, and attitudes required to perform a task effectively.
- **Senior Secondary Teachers:** Teachers who instruct students in upper secondary grades (typically Grades 10–13 in Sri Lanka).
- **Mathematics Achievement:** The level of students' performance in mathematics as measured by assessments.

Outline sections not yet in the candidate's draft (add if wanted): Hypotheses; Conceptual Framework (with figure); Ethical Considerations; Organisation of the Thesis.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

- ◆ **Theoretical Framework:** This section will review the relevant theories on attitudes and beliefs, including the Theory of Planned Behavior and Self-Efficacy Theory.
- ◆ **Previous Studies:** A review of previous research on the impact of teachers' attitudes and beliefs on instructional practices, specifically in mathematics education.
- ◆ **Problem-Solving Strategies in Mathematics:** An overview of effective problem-solving strategies and their importance in secondary education.
- ◆ **Contextual Factors in Sri Lanka:** A discussion on the educational context in Sri Lanka, including cultural, societal, and institutional influences on teaching practices.

This chapter reviews relevant literature related to the use of problem-solving strategies in mathematics education and the competency of teachers in implementing these strategies. It examines theoretical perspectives, empirical studies, and challenges associated with teaching problem-solving in mathematics, with special attention to the Sri Lankan context.

2.2 Concept of Problem Solving in Mathematics

Problem solving is a central component of mathematics learning. It involves the ability to apply mathematical knowledge and skills to unfamiliar situations in order to find solutions. According to George Pólya, problem solving is not merely about obtaining answers but understanding the process of arriving at solutions.

Pólya proposed four fundamental steps in problem solving:

1. Understanding the problem
2. Devising a plan
3. Carrying out the plan
4. Looking back (reflection)

These steps have been widely adopted in mathematics education and serve as a foundation for teaching problem-solving strategies.

2.3 Problem-Solving Strategies in Mathematics Teaching

Problem-solving strategies are structured approaches that help learners tackle mathematical problems effectively. Common strategies include:

- Drawing diagrams or visual representations
- Identifying patterns
- Working backward
- Making systematic lists
- Using logical reasoning

Research shows that students who are explicitly taught these strategies perform better in solving non-routine problems. Effective teaching of these strategies requires teachers to model thinking processes, encourage discussion, and create opportunities for exploration.

2.4 Teacher Competency in Mathematics Instruction

Teacher competency refers to the knowledge, skills, and attitudes required to facilitate effective learning. In mathematics education, this includes:

- Content knowledge
- Pedagogical knowledge
- Ability to use appropriate teaching strategies
- Classroom management skills

Competent teachers are able to adapt their teaching methods to suit students' needs and promote higher-order thinking skills. The integration of problem-solving strategies into teaching requires a deep understanding of both mathematics content and pedagogy.

2.5 Theoretical Framework

This study is grounded in constructivist learning theory, which emphasizes that learners actively construct knowledge through experience and interaction.

Key contributors to this theory include Jean Piaget and Lev Vygotsky.

- **Piaget's Theory:** Suggests that learners build knowledge through stages of cognitive development.
- **Vygotsky's Theory:** Highlights the importance of social interaction and the Zone of Proximal Development (ZPD), where learners can achieve higher understanding with guidance.

Problem-solving strategies align with these theories as they encourage active engagement, collaboration, and critical thinking.

2.6 Empirical Studies on Problem-Solving in Mathematics

Several studies have examined the impact of problem-solving strategies on mathematics learning:

- Studies indicate that students exposed to problem-solving-based instruction show improved conceptual understanding and retention.
- Research also shows that teacher training programs significantly enhance the use of problem-solving strategies in classrooms.
- Some studies highlight that teachers often lack sufficient training in implementing these strategies effectively.

In developing countries, including Sri Lanka, challenges such as limited resources, large class sizes, and exam-oriented teaching further hinder the use of problem-solving approaches.

2.7 Challenges in Implementing Problem-Solving Strategies

Teachers face several challenges when integrating problem-solving strategies into their teaching:

- **Lack of Professional Training:** Many teachers have not received adequate training in modern teaching methods.
- **Time Constraints:** Completing the syllabus often takes priority over in-depth problem-solving activities.
- **Examination Pressure:** Focus on high-stakes exams encourages rote learning.
- **Limited Resources:** Insufficient teaching materials and support.
- **Student Factors:** Students may lack foundational knowledge or motivation.

These challenges highlight the need for targeted interventions to enhance teacher competency.

2.8 Mathematics Education in Sri Lanka

In Sri Lanka, mathematics is a compulsory subject up to the General Certificate of Education (GCE) Ordinary Level. Despite curriculum reforms aimed at promoting student-centered learning, traditional teaching methods still dominate many classrooms.

The National Institute of Education Sri Lanka has introduced guidelines encouraging the use of activity-based and problem-solving approaches. However, implementation remains inconsistent due to gaps in teacher training and support.

2.9 Strategies for Enhancing Teacher Competency

Improving teacher competency requires a multifaceted approach:

- **Professional Development Programs:** Workshops and training sessions on problem-solving strategies.
- **Collaborative Learning:** Peer discussions and sharing best practices.
- **Use of Technology:** Incorporating digital tools to support teaching.
- **Reflective Practice:** Encouraging teachers to evaluate and improve their teaching methods.

Continuous professional development is essential for sustaining improvements in teaching practices.

2.10 Summary of the Literature Review

The literature highlights the importance of problem-solving strategies in mathematics education and the critical role of teacher competency in implementing these strategies effectively. While theoretical and empirical evidence supports the use of problem-solving approaches, challenges remain, particularly in contexts like Sri Lanka.

This study seeks to address these gaps by developing and evaluating an intervention aimed at enhancing teachers' competency in using problem-solving strategies.

Outline items not yet in the candidate's draft (add if wanted): Summary of related studies (table); Research Gap; Teacher Education Modules; G.C.E. O/L curriculum and O/L results evidence. The draft has no in-text citations yet; add them only for sources the candidate has read.

CHAPTER 3

METHODOLOGY

3.1 Introduction

Overview of the chapter and of the multi-phase mixed-methods design.

3.2 Research Paradigm

Philosophical stance (e.g. pragmatism) and why it fits a mixed-methods, intervention study.

3.3 Research Design

Objectives I, II, III, IV, V, VI

Multi-phase mixed-methods design (Proposal §4): Phase 1 quantitative survey → Phase 2 qualitative exploration → Phase 3 intervention → teacher education module. Explain how each phase informs the next.

Design diagram, redrawn from Proposal §4 (p. 8) after the sampling issues are resolved

Figure 3.1: Multi-phase mixed-methods research design

Table 3.1: Research phases and objectives, as stated in the proposal

Phase	Approach	Method	Objectives
Phase 1	Quantitative Phase	Cross-sectional survey	I, II
Phase 2	Qualitative Phase	Exploratory study	I, II, III
Phase 3	Quantitative & Qualitative mixed design using experimental design	Intervention (experimental / action component)	IV, V, VI

Phase 3 → IV, V, VI is inferred, not stated in the proposal. Confirm it.

3.4 Population

G.C.E. O/L (grades 10–11) mathematics teachers in Sri Lanka: number, distribution by province, zone and medium; inclusion criteria (currently teaching grade 10 or 11). The proposal says “secondary” in Participants but “senior secondary” in the title; state O/L explicitly.

3.5 Phase 1: Quantitative Survey

Objectives I, II

3.5.1 Sampling and Sample Size

One sampling frame and one N, with a justification (power analysis for regression; 5–10 respondents per item for factor analysis). The proposal gives 45 (stratified), 150 (convenience) and “purposive”; resolve to one.

Table 3.2: Phase 1 sample by education zone

Education zone	Schools	Teachers (population)	Teachers (sample)
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

3.5.2 Instrument: Teacher Questionnaire

Sections, number of items, response scale, and which items measure Objective I (cognitive ability) and which measure Objective II (level of use).

Table 3.3: Structure of the teacher questionnaire

Section	Construct	No. of items	Objective
—	—	—	—
—	—	—	—
—	—	—	—

Section	Construct	No. of items	Objective
—	—	—	—

3.5.3 Validity and Reliability

Expert panel and content validity index (CVI); pilot study (sample, setting); Cronbach's α ($\geq .70$) per scale; EFA/CFA if N allows.

3.5.4 Data Collection Procedure

Permissions (MOE, provincial and zonal offices), mode (online or in person), timing, follow-up.

3.5.5 Data Analysis

Software (R or Python), data screening, descriptive statistics, and the inferential tests for each research question. The proposal lists factor analysis, discriminant function analysis and regression.

3.6 Phase 2: Qualitative Exploration

Objectives I, II, III

3.6.1 Participants and Selection

How participants are drawn from Phase 1 (the proposal gives 45 and 30 interviews, and 20 observed classes). Selection criteria. Pseudonyms T01...

3.6.2 Instruments

Semi-structured interview schedule, classroom observation checklist, document analysis guide (lesson plans, schemes of work). Say how each was developed and piloted.

3.6.3 Data Collection Procedure

Interview setting, language, recording, transcription and translation; observation protocol.

3.6.4 Data Analysis

Reflexive thematic analysis (Braun & Clarke), coding in NVivo (the proposal names QSR N6, which is legacy software). Codebook development.

3.6.5 Trustworthiness

Credibility, transferability, dependability, confirmability; triangulation and member checking.

3.7 Phase 3: Intervention

Objectives IV, V

3.7.1 Design

Quasi-experimental pre/post design, with a comparison group if feasible. The proposal has no control group or pre/post instrument.

3.7.2 Participants

Sampling (the proposal says “convenient”), group sizes, assignment.

3.7.3 The Intervention

How Phase 1–2 findings shaped it; aims, content, duration, sessions, delivery mode, facilitators, materials.

Table 3.4: Intervention plan

Session	Focus / strategy	Activities	Duration
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

3.7.4 Fidelity of Implementation

Attendance, session checklists, observation and reflection logs.

3.7.5 Instruments and Measures

Pre/post measures of competency in using problem-solving strategies; classroom observation; teacher reflections; stakeholder feedback.

3.7.6 Data Analysis

Pre/post comparison (e.g. paired t-test or Wilcoxon; ANCOVA with a comparison group), effect sizes, qualitative analysis of reflections.

3.8 Development of the Teacher Education Module

Objective VI

The proposal has no method for Objective VI. Describe the design model, how findings feed into it, expert review and validation of the module.

3.9 Integration of Quantitative and Qualitative Data

Points of integration between phases; joint displays; meta-inferences.

3.10 Language of Instruments

Sinhala, Tamil and English versions; translation and back-translation procedure; checking equivalence.

3.11 Ethical Considerations

UoC Faculty of Education ethics clearance; permissions; information sheet and consent; the right to withdraw; confidentiality and pseudonyms; data storage; no identifiable data in AI tools.

3.12 Summary

A table or paragraph mapping objective → research question → phase → instrument → analysis.

Table 3.5: Alignment of objectives, questions, data and analysis

[illegible]

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

Objectives I, II, III, IV, V, VI

What this chapter reports and in what order: Phase 1 results, Phase 2 findings, Phase 3 results and the module, then the discussion.

4.2 Results of Phase 1: Quantitative Survey

4.2.1 Overview

Objectives I, II

What this chapter reports, and in what order (by research question).

4.2.2 Response Rate and Data Screening

Questionnaires distributed, returned and usable; missing data; outliers; normality checks.

4.2.3 Profile of Respondents

Table 4.1: Profile of respondents

Characteristic	Category	n	%
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

Gender, age, qualifications, professional training, years of O/L teaching, education zone, school type, medium.

4.2.4 Validity and Reliability of the Instrument

Table 4.2: Reliability of questionnaire scales

Scale	No. of items	Cronbach's α	Retained items
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

EFA/CFA results if run: KMO, Bartlett's test, loadings, fit indices.

4.2.5 Teachers' Cognitive Abilities Regarding Problem-Solving Strategies

Objective I

Table 4.3: Descriptive statistics for cognitive ability

Dimension	M	SD	Level
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

Results only; interpretation goes in the Discussion (§4.5).

4.2.6 Teachers' Level of Use of Problem-Solving Strategies

Objective II

Table 4.4: Descriptive statistics for level of use

Strategy	M	SD	Level
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

Chart built from the Phase 1 dataset

Figure 4.1: Level of use of problem-solving strategies

4.2.7 Differences by Teacher Characteristics

Objectives I, II

Group comparisons (e.g. by zone, experience, training), if they are among the research questions.

4.2.8 Relationships and Predictors

Objectives I, II

Table 4.5: Regression of level of use on predictors

Predictor	B	SE	β	p
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

Model summary (R^2 , F), assumptions checked.

4.2.9 Summary

Findings carried forward into Phase 2.

4.3 Findings of Phase 2: Qualitative Exploration

4.3.1 Overview

Objectives I, II, III

Purpose of Phase 2 and how it builds on Phase 1.

4.3.2 Profile of Participants

Table 4.6: Profile of Phase 2 participants

Pseudonym	Zone	School type	Years of O/L teaching	Data source
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

4.3.3 Analytic Process

Coding stages, codebook, and how themes were generated and reviewed.

Table 4.7: Themes and sub-themes

Theme	Sub-theme	Participants	Objective
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

4.3.4 Teachers' Understanding of Problem-Solving Strategies

Objective I

Themes on teachers' cognitive abilities, with anonymised quotations.

4.3.5 Teachers' Classroom Use of Problem-Solving Strategies

Objective II

4.3.5.1 Evidence from Interviews

Themes and quotations.

4.3.5.2 Evidence from Classroom Observation

Observation checklist results and field notes.

4.3.5.3 Evidence from Documents

Lesson plans, schemes of work, teachers' guides.

4.3.6 Challenges and Areas for Improvement

Objective III

Barriers, needs and areas for improvement identified by teachers and observed in class.

4.3.7 Integration with Phase 1 Results

Objectives I, II, III

Where the qualitative findings confirm, extend or contradict Phase 1.

Table 4.8: Joint display of Phase 1 and Phase 2 findings

Aspect	Phase 1 result	Phase 2 finding	Meta-inference
—	—	—	—
—	—	—	—
—	—	—	—

4.3.8 Summary

Findings that shape the intervention design in Phase 3.

4.4 Results of Phase 3: Intervention and Teacher Education Module

4.4.1 Overview

Objectives IV, V, VI

Purpose of Phase 3 and how Phases 1–2 shaped it.

4.4.2 Design of the Intervention

Objective IV

How the areas for improvement (Phase 2, §4.3) became the aims and content of the intervention; the final intervention plan.

Intervention flow diagram (candidate to design)

Figure 4.2: Structure of the intervention

4.4.3 Implementation of the Intervention

Objective IV

4.4.3.1 Participants and Setting

Table 4.9: Phase 3 participants

Group	n	Zone(s)	Schools
—	—	—	—
—	—	—	—
—	—	—	—

4.4.3.2 Delivery and Fidelity

Sessions delivered, attendance, deviations from the plan, fidelity checks.

4.4.4 Progress in the Use of Problem-Solving Strategies

Objective V

4.4.4.1 Pre-Intervention Results

Baseline measures.

4.4.4.2 Post-Intervention Results

Table 4.10: Pre- and post-intervention scores

Measure	Pre M (SD)	Post M (SD)	Test statistic	p	Effect size
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—

4.4.4.3 Classroom Observation after the Intervention

Changes observed in practice.

4.4.4.4 Teachers' Reflections and Stakeholder Feedback

Themes from reflections and feedback, with anonymised quotations.

4.4.5 Proposed Teacher Education Module

Objective VI

4.4.5.1 Rationale and Aims

Why the module is needed, drawing on all three phases.

4.4.5.2 Structure and Content

Table 4.11: Structure of the teacher education module

Unit	Learning outcomes	Content	Activities	Hours
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

4.4.5.3 Delivery and Assessment

Target users (pre-service or in-service), delivery mode, assessment of participants.

4.4.5.4 Validation of the Module

Expert review, pilot, revisions made.

4.4.6 Summary

Key results of Phase 3.

4.5 Discussion

4.5.1 Overview

How the discussion is organised: by objective, then integrated.

4.5.2 Teachers' Cognitive Abilities Regarding Problem-Solving Strategies

Objective I

Phase 1 and 2 findings set against the literature (Chapter 2).

4.5.3 Level of Use of Problem-Solving Strategies

Objective II

Discussion of Objective II findings.

4.5.4 Areas for Improvement

Objective III

Discussion of Objective III findings.

4.5.5 Effectiveness of the Intervention

Objectives IV, V

Discussion of Objectives IV and V.

4.5.6 The Teacher Education Module

Objective VI

How the module responds to the findings; comparison with existing programmes.

4.5.7 Integrated Discussion

Objectives I, II, III, IV, V, VI

Mixed-methods meta-inferences across the three phases.

4.5.8 Theoretical and Practical Contributions

What the study adds to the theory and practice of problem-solving teaching.

4.5.9 Summary

Key points of the discussion.

4.6 Summary

Key results and discussion points carried into Chapter 5.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

Purpose of the chapter.

5.2 Summary of the Study

Problem, objectives, design, sample and instruments in brief.

5.3 Major Findings

Objectives I, II, III, IV, V, VI

One subsection or paragraph per research question or objective.

5.4 Conclusions

What the findings mean in answer to each research question.

5.5 Recommendations

5.5.1 For Mathematics Teachers

Candidate to draft.

5.5.2 For Teacher Educators and Teacher Education Institutions

NIE, National Colleges of Education, universities, in-service advisers.

5.5.3 For Policy Makers

Ministry of Education, provincial and zonal education offices.

5.6 Contributions of the Study

Contribution to knowledge, practice and policy. Source: Proposal §5.

5.7 Limitations

Limitations that affect how far the findings can be generalised.

5.8 Suggestions for Further Research

Candidate to draft.

5.9 Summary

Closing paragraph.

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APPENDICES

Appendix-1: Ethics Clearance

UoC Faculty of Education ethics approval letter.

Appendix-2: Permission Letters

Ministry of Education, provincial and zonal education offices, principals.

Appendix-3: Participant Information Sheet and Consent Form

Blank templates only, in Sinhala, Tamil and English. No signed forms.

Appendix-4: Teacher Questionnaire (Phase 1)

Final version in each language; note the translation procedure.

Appendix-5: Content Validity: Expert Panel and CVI

Panel composition (anonymised) and item-level CVI.

Appendix-6: Interview Schedule (Phase 2)

Candidate to draft.

Appendix-7: Classroom Observation Checklist

Candidate to draft.

Appendix-8: Document Analysis Guide

Candidate to draft.

Appendix-9: Pre- and Post-Intervention Instruments (Phase 3)

Candidate to draft.

Appendix-10: Intervention Plan and Session Materials

Candidate to draft.

Appendix-11: Teacher Education Module

Full module document.